

MARTHA'S VINEYARD, MA, USA

WMO: 725066

Lat: 41.393N

Lon: 70.615W

Elev: 21

StdP: 101.08

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 94724

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.8	-21.4	0.6	-9.8	-18.8	0.7	-7.9	14.0	-0.2	12.5	0.0	4.6	330	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	29.0	22.7	27.6	22.0	26.3	21.4	24.3	27.2	23.6	26.1	22.8	25.0	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.1	25.9	22.7	17.4	25.0	22.2	16.9	24.4	73.2	27.4	70.2	26.2	67.2	25.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.6	10.5	9.1	DB	-16.7	31.9	3.3	1.6	-19.1	33.0	-21.0	34.0	-22.8	34.9	-25.2	36.0
(5)				WB	-17.3	25.6	3.1	0.9	-19.5	26.3	-21.3	26.8	-23.1	27.3	-25.4	27.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-0.2	0.0	3.0	7.8	13.1	17.6	21.3	20.8	17.3	12.0	7.3	3.0
	DBStd	8.58	5.44	4.48	4.31	3.50	3.55	3.35	3.01	3.12	3.70	4.28	4.58	4.80
	HDD10.0	1272	318	281	219	83	11	0	0	0	1	29	106	224
	HDD18.3	3192	576	514	476	316	165	52	9	12	62	201	332	477
	CDD10.0	1382	1	1	3	18	108	228	349	336	221	90	24	6
	CDD18.3	261	0	0	0	1	4	31	100	90	32	4	0	0
	CDH23.3	1364	0	0	0	6	27	150	549	522	105	6	0	0
	CDH26.7	237	0	0	0	2	4	29	111	82	7	1	0	0
Wind	WSAvg	4.5	5.0	5.0	5.3	5.0	4.5	4.0	3.7	3.5	3.9	4.5	4.8	4.9
Precipitation	PrecAvg	1137	86	81	113	97	90	85	69	90	101	99	103	118
	PrecMax	1505	150	157	219	184	166	264	157	196	327	223	192	234
	PrecMin	991	42	41	20	37	26	11	12	3	22	17	19	38
	PrecStd	111	28	31	51	42	37	53	31	49	62	55	46	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	12.4	16.6	21.3	26.7	29.8	31.5	30.9	27.5	23.7	18.6	16.1
		MCWB	11.5	9.4	11.0	13.0	18.5	22.4	24.4	23.7	21.6	19.4	15.8	14.2
	2%	DB	11.4	10.3	13.0	17.4	23.1	27.1	29.3	28.6	25.9	21.7	17.3	14.1
		MCWB	10.5	8.5	8.7	11.1	16.8	20.8	23.0	22.9	21.1	18.6	15.2	12.6
	5%	DB	9.9	8.8	11.2	15.2	21.2	24.9	27.7	27.4	24.4	20.1	16.1	12.7
		MCWB	8.5	7.0	8.2	10.4	15.7	19.2	22.3	22.3	20.2	17.5	13.9	11.3
	10%	DB	7.9	7.2	9.5	13.6	19.0	23.1	26.3	26.2	23.1	18.8	14.6	11.1
		MCWB	6.2	5.0	7.2	9.5	14.5	18.8	21.6	21.7	19.6	16.6	12.7	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	10.8	12.9	15.4	19.9	23.5	25.5	25.6	23.9	21.0	17.2	14.7
		MCDB	12.6	11.2	15.0	19.0	24.5	28.0	30.0	28.8	26.4	22.2	17.8	15.5
	2%	WB	10.7	8.9	10.6	13.4	18.2	21.9	24.3	24.6	22.7	19.5	15.9	13.1
		MCDB	11.2	10.0	11.9	15.9	21.8	25.3	27.5	27.1	24.4	20.9	16.8	14.0
	5%	WB	8.7	7.4	9.0	11.9	16.8	20.9	23.5	23.7	21.8	18.3	14.5	11.7
		MCDB	9.5	8.3	10.5	13.4	19.7	23.6	26.3	26.0	23.2	19.5	15.8	12.5
	10%	WB	6.5	5.3	7.6	10.8	15.5	19.8	22.7	22.9	20.9	17.1	13.0	9.7
		MCDB	7.7	6.9	9.0	12.6	18.0	22.1	25.1	25.0	22.4	18.4	14.4	10.9
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	8.6	9.5	9.5	9.1	9.0	9.6	10.3	9.8	8.8	8.4
		MCDBR	9.0	10.0	10.2	12.5	12.2	11.3	10.0	9.7	10.3	10.1	8.6	9.0
		MCWBR	8.8	8.8	7.7	7.6	6.8	6.4	5.2	5.4	6.8	7.5	7.6	9.0
	5% WB	MCDBR	8.9	9.8	9.0	9.0	10.3	8.8	7.8	7.5	7.5	8.2	7.8	8.7
		MCWBR	9.1	9.5	8.1	6.8	6.3	5.8	4.6	4.6	5.6	7.1	7.9	9.2
Clear-Sky Solar Irradiance	taub	0.312	0.318	0.342	0.380	0.418	0.466	0.491	0.445	0.378	0.351	0.322	0.310	
	taud	2.502	2.490	2.459	2.377	2.312	2.215	2.153	2.277	2.440	2.520	2.559	2.552	
	Ebn at Noon	844	895	909	891	860	816	791	819	858	846	829	819	
	Edh at Noon	74	87	100	117	128	141	149	127	101	84	70	66	
All-Sky Solar Radiation	RadAvg	1.79	2.69	3.73	4.81	5.40	5.82	5.98	5.38	4.38	2.99	2.02	1.52	
	RadStd	0.14	0.27	0.29	0.51	0.52	0.54	0.32	0.30	0.32	0.26	0.14	0.15	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	+48

Nomenclature: See separate page